DBSCAN

KMEANS v/s DBSCAN:

⇒ In K-Means there is a need to specify the no. of clusters initially.

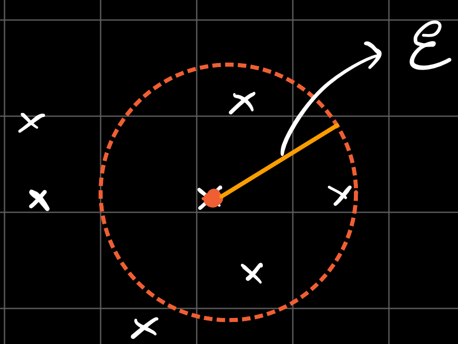
⇒ K-Means is very sensitive to outliers.

* ⇒ K-Means is a centriod based algo. which dosent work on spherical data.

DBSCAN: (Density Based Spatial Clustering of Application with Noise)

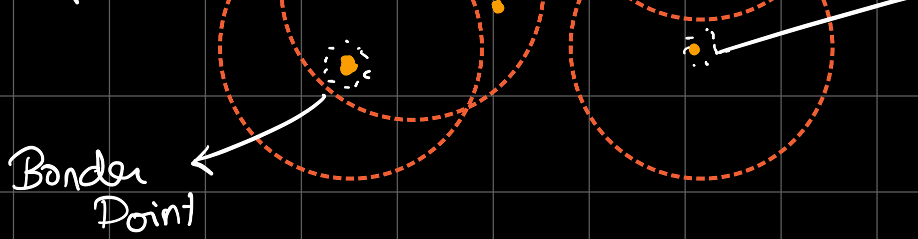
$\varepsilon \rightarrow$ unit length

minpoints $\rightarrow 3$



If no. of points > minpoints,
density = dense

If no. of points < minpoints,
density = scarce



TYPES OF POINTS:

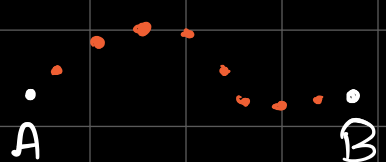
1) CORE POINT: Point with no of points > min point in ε radius.

2) BORDER POINT: Point with no of points < min point and a core point in ε radius.

3) NOISE POINTS: Point with no of points < min point and no core point in ε radius.

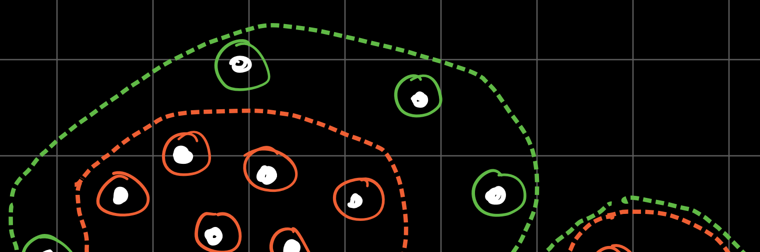
DENSITY CONNECTED POINTS:

If two points are density connected they can be a part of the same cluster.



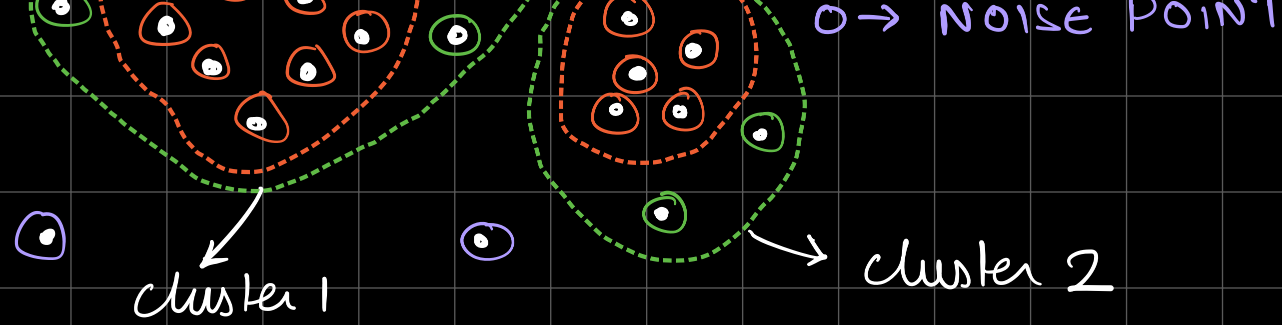
⇒ If A & B are indirectly connected by core points with a distance less than ε b/w the core points they are said to be density connected.

Ex:



O → CORE POINT

O → BORDER POINT



Step 1) Identify all points as core, boundary & noise.

Step 2) For all new unclustered core points create a cluster by using density connections.

Step 3) For border points add them to the same cluster as the nearest core point.

Step 4) Leave all noise point as its is.